

Digital Electronics and Pulse Techniques

Assignment-04

Total – 20 marks

Question 1

03

For the NMOS given in the figure,

M1 and M3: $K_n = 0.3 \text{ mA/V}^2$ $V_{TN} = 0.6 \text{ V}$

M2: $K_n = 0.4 \text{ mA/V}^2$, $V_{TN} = 0.7 \text{ V}$

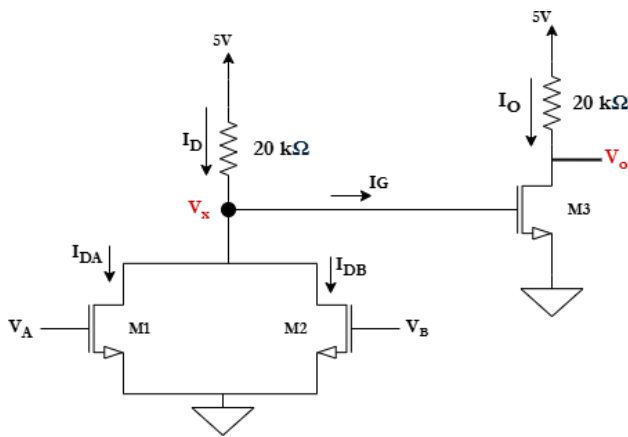


Figure 1

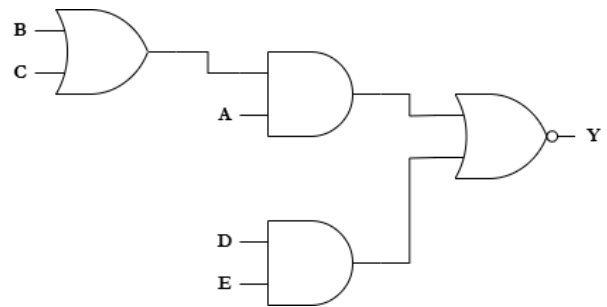


Figure 2

(a)	From the above Figure 1, determine the currents I_D , I_{DA} , I_{DB} , I_O , V_X , and V_O for the cases where both inputs are high and both inputs are low. (Consider 5 V as logic high and 0 V as logic low)	1
(b)	Determine the power dissipations for the cases where both inputs are high and both inputs are low.	1
(c)	Find the Boolean expression for output Y from Figure 2 and, using that expression, design an equivalent CMOS logic circuit of Figure 2.	1

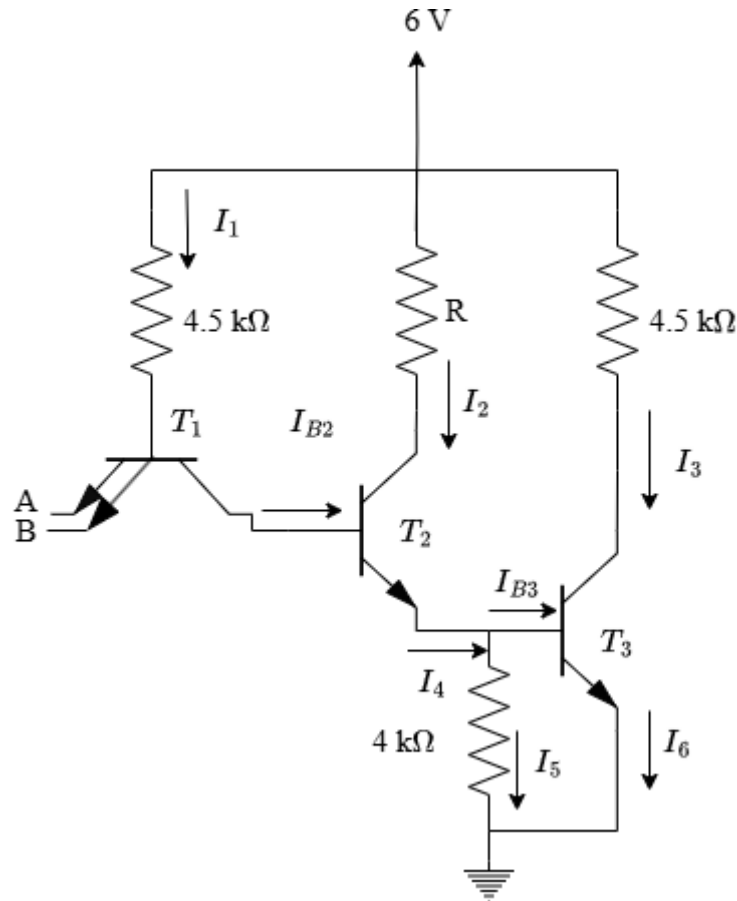
Question 2

02

For the given TTL NAND gate,

$\beta_R = 0.2$, $V_{CE}(\text{saturation}) = 0.2 \text{ V}$, $V_{BE}(\text{saturation}) = 0.8 \text{ V}$, $V_{BC}(\text{reverse active}) = 0.7 \text{ V}$.

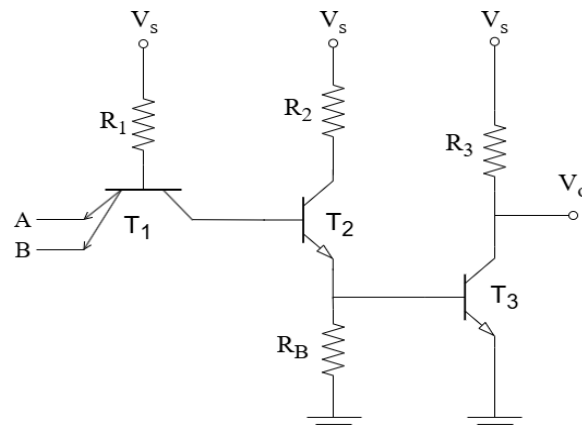
$V_{in}(\text{high}) = 6 \text{ V}$, $V_{in}(\text{low}) = 0.2 \text{ V}$



(a)	If $I_6 = 4.74 \text{ mA}$ when both inputs are high , find the value of R .	1
(b)	Determine the power consumption for the above case.	1

Question 3

02

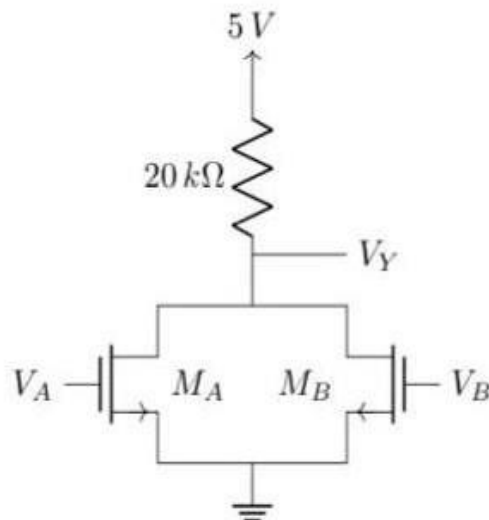


$V_s = 9\text{ V}$, $R_2 = 2\text{ k}$, $R_3 = 4.4\text{ k}$, $R_B = 4.4\text{ k}$

(a)	Determine the power dissipation of this TTL circuit when both inputs are HIGH. You must show your calculation. Assume $R_1 = 6.7\text{ k}$ for this question only	1
(b)	Suppose you are selling this TTL logic gate to a customer, and he states a strict criterion to keep the power dissipation below 1.23 mW when both inputs are LOW (0 V). Determine the maximum value of R_1 you can use to satisfy this criterion.	1

Question 4

03

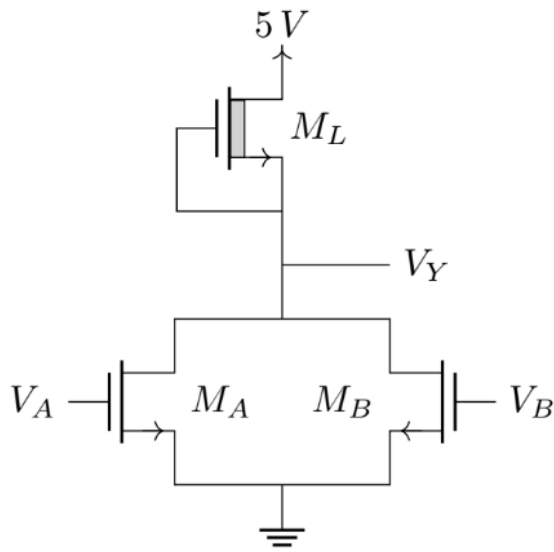


For the NOR gate shown, $V_{TN_A} = 0.25\text{ V}$, $V_{TN_B} = 0.3\text{ V}$, $K_{n_A} = 0.4 \frac{\text{A}}{\text{V}^2}$, $K_{n_B} = 0.5 \frac{\text{A}}{\text{V}^2}$.

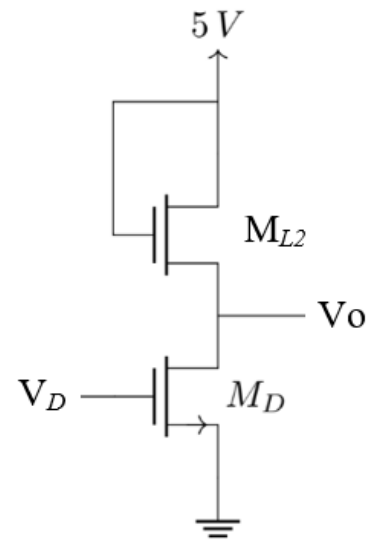
(a)	Find the lower threshold of output voltage.
(b)	Find the average power dissipation.
(c)	If a new NMOS M_C is used instead of M_B such that the drain currents of M_A & M_C are equal when both inputs are high, find the value of K_{n_C} . It is given that $V_{TN_C} = V_{TN_B} = 0.3\text{ V}$.

Question 5

03



Circuit - 1



Circuit - 2

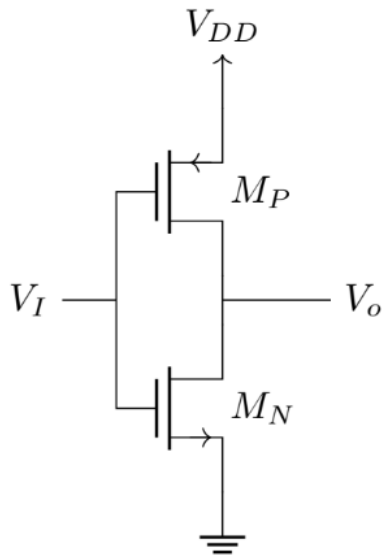
For Circuit – 1, It is given that $V_{TA} = V_{TB} = 0.5 \text{ V}$, $V_{TL} = -0.6 \text{ V}$, $K_{nA} = K_{nB} = 10 \text{ mA/V}^2$ & $K_{nL} = 30 \text{ mA/V}^2$

For Circuit – 2, It is given that $V_{TD} = 1 \text{ V}$, $V_{TL} = 0.8 \text{ V}$, $K_{nD} = 50 \text{ mA/V}^2$ & $K_{nL2} = 10 \text{ mA/V}^2$

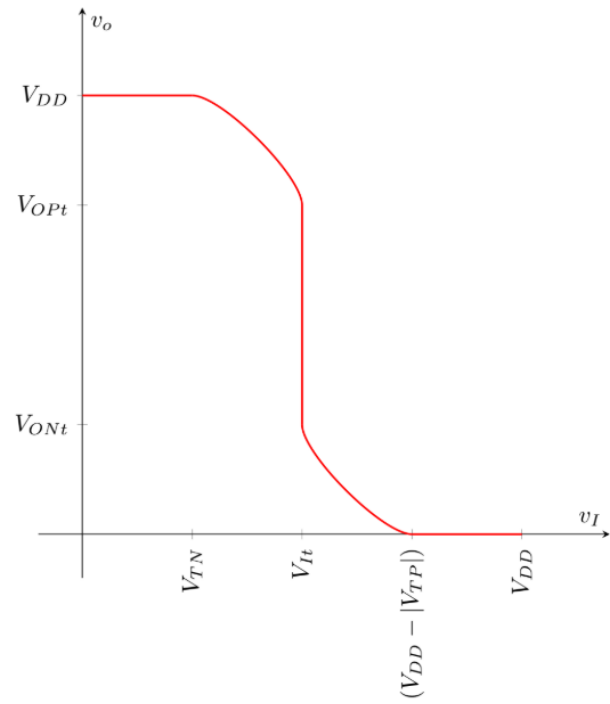
(a)	Find average power dissipation for Circuit-1.	1
(b)	If V_Y is connected to V_D , find the values of V_O and total power consumption of the combined circuit when – i) $V_A = V_B = 0 \text{ V}$ & ii) $V_A = V_B = 5 \text{ V}$	2

Question 6

02



CMOS Inverter



Ideal VTC of a CMOS Inverter

It is given that $V_{DD} = 5 \text{ V}$, $V_{TN} = -V_{TP} = 0.5 \text{ V}$, $K_N = K_P = 10 \mu\text{A/V}^2$

(a)	Derive the equations of critical voltages V_{It} , V_{ONt} , and V_{OPt} and then find their values	1
(b)	Determine the noise margin of the above CMOS Inverter	1

Question 7

05

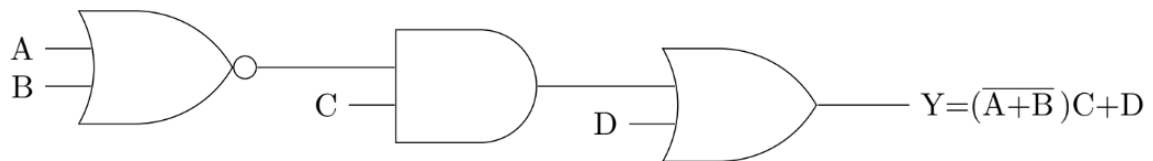
Implement the following Boolean logic using a CMOS logic circuit –

i) $\bar{A} \cdot \bar{B} \cdot \bar{C}$

ii) $\overline{AB} + C$

iii) $(AB+C)D$

iv)



v)

